

LEESHA R

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LAB 3 — ARCHITECTURE SELECTION & WRITTEN JUSTIFICATION

System: Pet Adoption & Shelter Management System

Architectural Style: Layered Architecture

Architecture Selection

We chose **Layered Architecture** for the **Pet Adoption & Shelter Management System**.

Architectural Choice

The system is organized into separate layers and components for the user interface, application/service processing, and data management. The **Adoption Portal / UI** provides the interface for adoption applicants, while the **Order Manager**, **Pet & Shelter Management**, and **Payment Service** components handle the main application and service operations. The **Shelter Database** forms the data storage component.

Reason 1 — Separation of Concerns

Layered Architecture separates different responsibilities of the system. The Adoption Portal / UI can handle applicant interactions such as searching available pets, viewing pet details, and submitting adoption applications without directly handling database operations. The Order Manager and Pet & Shelter Management components can independently handle application processing and pet/shelter information.

Reason 2 — Easier Maintenance and Data Management

The separation of components makes the system easier to understand, maintain, and modify. For example, changes to pet availability or shelter information can be handled by the Pet & Shelter Management component without requiring major changes to the user interface. Adoption and payment operations can also be handled through their respective service components.

Security Advantage

The architecture allows security and validation checks to be handled in the service layer before requests access sensitive data or payment functionality. This helps prevent the user interface from directly accessing the Shelter Database and allows access control and input validation to be applied consistently.

Performance Benefit

Separating the presentation, service, and data responsibilities allows individual components to process their tasks efficiently. Frequently requested pet and shelter information can be handled by the appropriate service component, while database operations remain separated from the user interface, reducing unnecessary direct database access.